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ELITE IAL MATHEMATICS

EDEXCEL INTERNATIONAL A LEVEL

WMA11/01 January 2023 Worked Solutions

Jan 2023 paper rebuilt with the worked solution after each question.

11
paper questions

WMA11
Pure 1

Single-paper solutions
standalone IAL route

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Prepared for Dr Eslam Ahmed - eliteigcse.com

P1

PAST PAPER

WMA11/01 January 2023

Jan 2023 | 11 questions | 75 marks

11
questions

75
marks

Answers
worked solution after each question

WMA11/01 JANUARY 2023

5 marks

Question 1

Differentiation

1. A curve C has equation

$$y = 2 + 10x^{\frac{1}{2}} - 2x^{\frac{3}{2}} \qquad\qquad x > 0$$

(a) Find $\frac{dy}{dx}$ giving your answer in simplest form.

(3)

(b) Hence find the exact value of the gradient of the tangent to C at the point where $x = 2$ giving your answer in simplest form.

(*Solutions relying on calculator technology are not acceptable*.)

(2)

(Total for Question 1 is 5 marks)

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Worked Solution - Question 1

Differentiation

Primary topic

1. Differentiate term by term

$y = 2 + 10x^{1/2} - 2x^{3/2}$, so $\dfrac{dy}{dx} = 5x^{-1/2} - 3x^{1/2}$.

2. Write in simplest form

$$\frac{dy}{dx} = \frac{5}{\sqrt{x}} - 3\sqrt{x}.$$

3. Substitute x = 2

At $x = 2$, the gradient is $\dfrac{5}{\sqrt{2}} - 3\sqrt{2} = \dfrac{5}{\sqrt{2}} - \dfrac{6}{\sqrt{2}} = -\dfrac{1}{\sqrt{2}}$.

4. Rationalise

$$-\frac{1}{\sqrt{2}} = -\frac{\sqrt{2}}{2}.$$

Final answer

$(a)\ \dfrac{dy}{dx} = \dfrac{5}{\sqrt{x}} - 3\sqrt{x}.$

$(b)\ -\dfrac{\sqrt{2}}{2}.$

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Question 2

5 marks

Straight Line

2. The points P, Q and R have coordinates $(-3, 7)$, $(9, 11)$ and $(12, 2)$ respectively.

(a) Prove that angle $PQR = 90°$

(3)

Given that the point S is such that $PQRS$ forms a rectangle,

(b) find the coordinates of S.

(2)

(Total for Question 2 is 5 marks)

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Worked Solution - Question 2

Straight Line

Primary topic

1. Find the gradient of PQ

$$m_{PQ} = \frac{11-7}{9-(-3)} = \frac{4}{12} = \frac{1}{3}.$$

2. Find the gradient of QR

$$m_{QR} = \frac{2-11}{12-9} = \frac{-9}{3} = -3.$$

3. Prove the right angle

$m_{PQ}m_{QR} = \frac{1}{3}(-3) = -1$, so PQ and QR are perpendicular. Hence angle $PQR = 90^\circ$.

4. Use the rectangle vector

For rectangle $PQRS$, $\overrightarrow{PS} = \overrightarrow{QR} = (12-9, 2-11) = (3, -9)$.

5. Find S

$$S = P + \overrightarrow{QR} = (-3, 7) + (3, -9) = (0, -2).$$

Final answer

(a) Angle $PQR = 90^\circ$.

(b) $S = (0, -2)$.

Question 3

5 marks

Integration

3. Find

$$\int \frac{4x^5 + 3}{2x^2} \, dx$$

giving your answer in simplest form.

(5)

(Total for Question 3 is 5 marks)

Worked Solution - Question 3

Integration

Primary topic

1. Split the fraction

$$\frac{4x^5+3}{2x^2} = 2x^3 + \frac{3}{2}x^{-2}.$$

2. Integrate the first term

$$\int 2x^3 dx = \frac{x^4}{2}.$$

3. Integrate the second term

$$\int \frac{3}{2}x^{-2} dx = -\frac{3}{2}x^{-1}.$$

4. Write in simplest form

The integral is $\frac{x^4}{2} - \frac{3}{2x} + c.$

Final answer

$$\frac{x^4}{2} - \frac{3}{2x} + c.$$

Question 4

4 marks

Quadratics

4. Given that the equation

$$kx^2 + 6kx + 5 = 0 \qquad \text{where } k \text{ is a non zero constant}$$

has no real roots, find the range of possible values for k.

(4)

(Total for Question 4 is 4 marks)

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Worked Solution - Question 4

Quadratics

Primary topic

1. Use the discriminant condition

For no real roots, the discriminant must be negative: $b^2 - 4ac < 0$.

2. Substitute a, b, and c

For $kx^2 + 6kx + 5 = 0$, $a = k$, $b = 6k$, and $c = 5$.

3. Form the inequality

$(6k)^2 - 4(k)(5) < 0$, so $36k^2 - 20k < 0$.

4. Solve the inequality

$4k(9k - 5) < 0$, so the product is negative when $0

Final answer

$0

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Question 5

6 marks

Indices & Surds

5. **In this question you must show all stages of your working.**

Solutions relying on calculator technology are not acceptable.

(a) By substituting $p = 3^x$, show that the equation

$$3 \times 9^x + 3^{x+2} = 1 + 3^{x-1}$$

can be rewritten in the form

$$9p^2 + 26p - 3 = 0$$

(3)

(b) Hence solve

$$3 \times 9^x + 3^{x+2} = 1 + 3^{x-1}$$

(3)

(Total for Question 5 is 6 marks)

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Worked Solution - Question 5

Indices & Surds

Primary topic

1. Let p = 3^x

Then $9^x = 3^{2x} = p^2$, $3^{x+2} = 9p$, and $3^{x-1} = \frac{p}{3}$.

2. Substitute into the equation

$3p^2 + 9p = 1 + \frac{p}{3}$.

3. Clear the fraction

Multiplying by 3 gives $9p^2 + 27p = 3 + p$, hence $9p^2 + 26p - 3 = 0$.

4. Solve for p

$9p^2 + 26p - 3 = (9p - 1)(p + 3) = 0$, so $p = \frac{1}{9}$ or $p = -3$.

5. Reject the negative value

Since $p = 3^x > 0$, we reject $p = -3$. Therefore $3^x = \frac{1}{9} = 3^{-2}$.

6. Find x

$x = -2$.

Final answer

$x = -2$.

Question 6

10 marks

Radians

6.

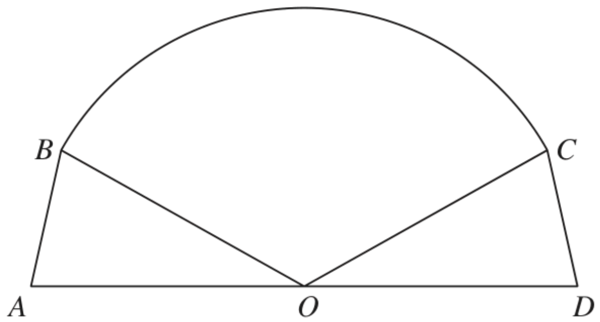

Diagram **NO** accurately dr

Figure 1

Figure 1 shows the plan view for the design of a stage.

The design consists of a sector OBC of a circle, with centre O, joined to two congruent triangles OAB and ODC.

Given that

- angle $BOC = 2.4$ radians
- area of sector $BOC = 40\,\text{m}^2$
- AOD is a straight line of length 12.5 m

(a) find the radius of the sector, giving your answer, in m, to 2 decimal places,

(2)

(b) find the size of angle AOB, in radians, to 2 decimal places.

(1)

Hence find

(c) the total area of the stage, giving your answer, in m^2, to one decimal place,

(3)

(d) the total perimeter of the stage, giving your answer, in m, to one decimal place.

(4)

(Total for Question 6 is 10 marks)

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Worked Solution - Question 6

Radians

Primary topic

1. Find the radius

Area of sector BOC is $\frac{1}{2}r^2\theta$. So $40 = \frac{1}{2}r^2(2.4) = 1.2r^2$, giving $r = \sqrt{\frac{40}{1.2}} = 5.77$ m.

2. Find angle AOB

The triangles OAB and ODC are congruent, so $\angle AOB = \angle COD$. Since AOD is a straight line, $2\angle AOB + 2.4 = \pi$.

3. Calculate the angle

$\angle AOB = \frac{\pi - 2.4}{2} = 0.3708\ldots$, so 0.37 rad.

4. Find the total area

The two congruent triangle areas together are $2\left(\frac{1}{2}(6.25)(5.7735\ldots)\sin 0.3708\ldots\right) = 13.075\ldots$.

5. Add the sector area

Total area $= 40 + 13.075\ldots = 53.1\ \mathrm{m}^2$ to one decimal place.

6. Find AB

Using the cosine rule, $AB^2 = 6.25^2 + r^2 - 2(6.25)(r)\cos 0.3708\ldots$, giving $AB = 2.265\ldots$ m.

7. Find the perimeter

Perimeter $= AD + AB + CD + \text{arc } BC = 12.5 + 2(2.265\ldots) + 5.7735\ldots(2.4) = 30.9$ m.

Final answer

(a) $r = 5.77$ m.

(b) 0.37 rad.

(c) $53.1\ \mathrm{m}^2$.

(d) 30.9 m.

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10 marks

Question 7

Transformations

7. (a) On Diagram 1, sketch a graph of the curve C with equation

$$y = \frac{6}{x} \qquad x \neq 0$$

(2)

The curve C is transformed onto the curve with equation $y = \dfrac{6}{x-2} \qquad x \neq 2$

(b) Fully describe this transformation.

(2)

The curve with equation

$$y = \frac{6}{x-2} \qquad x \neq 2$$

and the line with equation

$$y = kx + 7 \qquad \text{where } k \text{ is a constant}$$

intersect at exactly two points, P and Q.

Given that the x coordinate of point P is -4

(c) find the value of k,

(2)

(d) find, using algebra, the coordinates of point Q.

(*Solutions relying entirely on calculator technology are not acceptable.*)

(4)

Question 7 continued

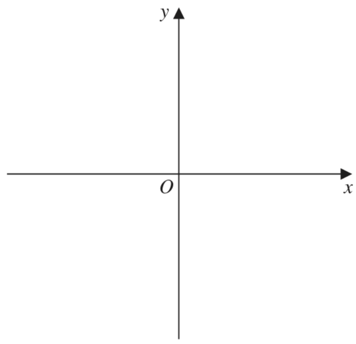
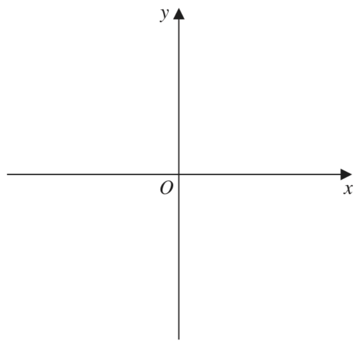


Diagram 1

Only use this copy of Diagram 1 if you need to redraw your graph.



Copy of Diagram 1

(Total for Question 7 is 10 marks)

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Worked Solution - Question 7

Transformations

Primary topic

1. Sketch y = 6/x

The graph is a rectangular hyperbola with asymptotes $x = 0$ and $y = 0$, with branches in quadrants I and III.

2. Describe the transformation

$y = \dfrac{6}{x-2}$ is obtained from $y = \dfrac{6}{x}$ by a translation 2 units in the positive x-direction.

3. Use P to find k

At $x = -4$ on $y = \dfrac{6}{x-2}$, $y = \dfrac{6}{-6} = -1$.

4. Substitute into the line

Since P lies on $y = kx + 7$, $-1 = -4k + 7$, so $k = 2$.

5. Find the second intersection

Solve $\dfrac{6}{x-2} = 2x + 7$. Multiplying by $x - 2$ gives $6 = (2x + 7)(x - 2)$.

6. Factor and choose Q

$6 = 2x^2 + 3x - 14$, so $2x^2 + 3x - 20 = 0 = (2x - 5)(x + 4)$. The root $x = -4$ is P, so Q has $x = \dfrac{5}{2}$.

7. Find y for Q

$y = 2\left(\dfrac{5}{2}\right) + 7 = 12$, so $Q = \left(\dfrac{5}{2}, 12\right)$.

Final answer

(b) Translation 2 units right.

(c) $k = 2$.

(d) $Q = \left(\frac{5}{2}, 12\right)$.

Question 8

8 marks

Quadratics

8.

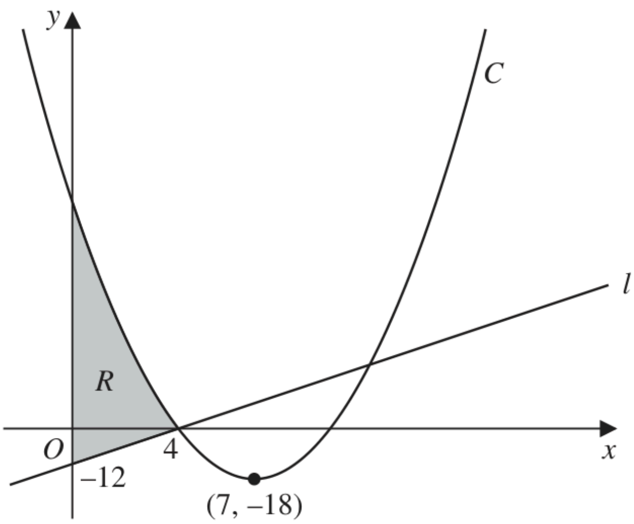

Figure 2

Figure 2 shows a sketch of the straight line l and the curve C.

Given that l cuts the y-axis at –12 and cuts the x-axis at 4, as shown in Figure 2,

(a) find an equation for l, writing your answer in the form $y = mx + c$, where m and c are constants to be found.

(2)

Given that C

- has equation $y = \mathrm{f}(x)$ where $\mathrm{f}(x)$ is a quadratic expression
- has a minimum point at (7, –18)
- cuts the x-axis at 4 and at k, where k is a constant

(b) deduce the value of k,

(1)

(c) find $\mathrm{f}(x)$.

(3)

The region R is shown shaded in Figure 2.

(d) Use inequalities to define R.

(2)

(Total for Question 8 is 8 marks)

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Worked Solution - Question 8

Quadratics

Primary topic

1. Find the straight line

The line cuts the y-axis at -12 and the x-axis at 4, so its gradient is $\dfrac{0-(-12)}{4-0} = 3$. Hence l is $y = 3x - 12$.

2. Use symmetry of the quadratic

The minimum point has x-coordinate 7. The roots are 4 and k, so their midpoint is 7.

3. Find k

$\dfrac{4+k}{2} = 7$, so $k = 10$.

4. Build the quadratic

Let $f(x) = a(x-4)(x-10)$. Since the minimum point is $(7, -18)$, $-18 = a(3)(-3) = -9a$, so $a = 2$.

5. Write f(x)

$f(x) = 2(x-4)(x-10) = 2x^2 - 28x + 80$.

6. Identify the shaded region

The region lies between the y-axis and $x = 4$, above the line and below the curve: $0 \leq x \leq 4$ and $3x - 12 \leq y \leq 2x^2 - 28x + 80$.

Final answer

(a) $y = 3x - 12.$

(b) $k = 10.$

(c) $f(x) = 2(x - 4)(x - 10).$

(d) $0 \leq x \leq 4,\ 3x - 12 \leq y \leq 2x^2 - 28x + 80.$

9.

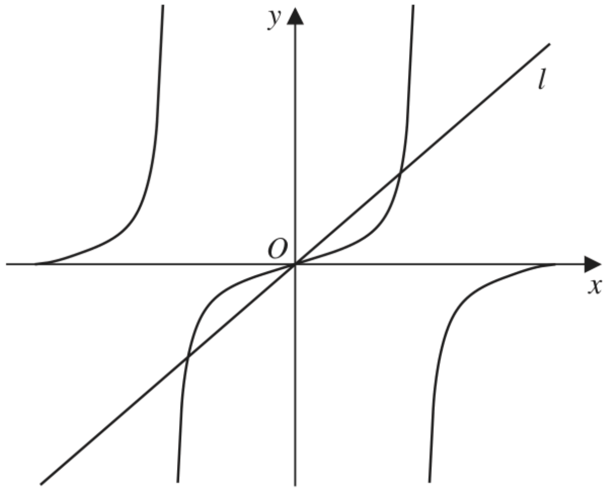

Figure 3

Figure 3 shows a sketch of

- the curve with equation $y = \tan x$
- the straight line l with equation $y = \pi x$

in the interval $-\pi < x < \pi$

(a) State the period of $\tan x$

(1)

(b) Write down the number of roots of the equation

(i) $\tan x = (\pi + 2)x$ in the interval $-\pi < x < \pi$

(1)

(ii) $\tan x = \pi x$ in the interval $-2\pi < x < 2\pi$

(1)

(iii) $\tan x = \pi x$ in the interval $-100\pi < x < 100\pi$

(1)

(Total for Question 9 is 4 marks)

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Worked Solution - Question 9

Trigonometric Functions

Primary topic

1. State the period

The period of $\tan x$ is π.

2. Count roots for the steeper line

In $-\pi

3. Extend to -2pi < x < 2pi

For $y = \pi x$, the interval $-2\pi

4. Extend to -100pi < x < 100pi

There are 99 positive outer roots, 99 negative outer roots, and 3 central roots, so the total is $99 + 99 + 3 = 201$.

Final answer

$(a)\ \pi.$

$(b)(i)\ 3,\ (ii)\ 5,\ (iii)\ 201.$

10.

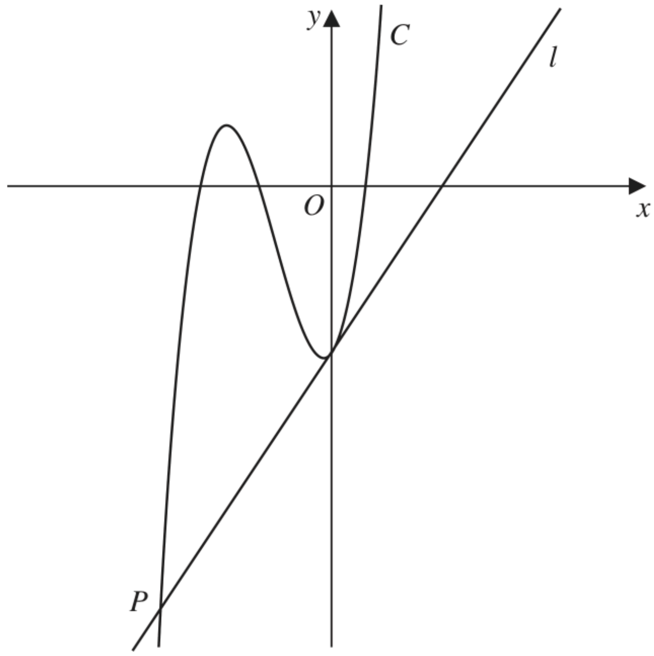

Figure 4

Figure 4 shows a sketch of part of the curve C with equation $y = \mathrm{f}(x)$, where

$$\mathrm{f}(x) = (3x + 20)(x + 6)(2x - 3)$$

(a) Use the given information to state the values of x for which

$$\mathrm{f}(x) > 0$$

(2)

(b) Expand $(3x + 20)(x + 6)(2x - 3)$, writing your answer as a polynomial in simplest form.

(3)

The straight line l is the tangent to C at the point where C cuts the y-axis.

Given that l cuts C at the point P, as shown in Figure 4,

(c) find, using algebra, the x coordinate of P

(*Solutions based on calculator technology are not acceptable*.)

(5)

(Total for Question 10 is 10 marks)

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Worked Solution - Question 10

Differentiation

Primary topic

1. Use the roots

$f(x) = (3x+20)(x+6)(2x-3)$ has roots $x = -\dfrac{20}{3}$, $x = -6$, and $x = \dfrac{3}{2}$.

2. State where f(x) is positive

Since the cubic has positive leading coefficient, $f(x) > 0$ for $-\dfrac{20}{3}\dfrac32$.

3. Expand the first two brackets

$(3x+20)(x+6) = 3x^2 + 38x + 120$.

4. Complete the expansion

$(3x^2+38x+120)(2x-3) = 6x^3 + 67x^2 + 126x - 360$.

5. Find the tangent at the y-axis

$f'(x) = 18x^2 + 134x + 126$, so at $x = 0$ the tangent gradient is 126. Also $f(0) = -360$.

6. Write the tangent equation

The tangent line is $y = 126x - 360$.

7. Find the other intersection

Set $6x^3 + 67x^2 + 126x - 360 = 126x - 360$. This gives $6x^3 + 67x^2 = 0$, so $x^2(6x+67) = 0$.

8. Choose point P

$x = 0$ is the tangent point on the y-axis, so the other intersection P has $x = -\dfrac{67}{6}$.

Final answer

$(a)\ -\dfrac{20}{3}\dfrac32.$

$(b)\ 6x^3 + 67x^2 + 126x - 360.$

$(c)\ x_P = -\dfrac{67}{6}.$

Question 11

8 marks

Integration

11. A curve C has equation $y = \mathrm{f}(x), \quad x > 0$

Given that

- $\mathrm{f}''(x) = 4x + \dfrac{1}{\sqrt{x}}$
- the point P has x coordinate 4 and lies on C
- the tangent to C at P has equation $y = 3x + 4$

(a) find an equation of the normal to C at P

(2)

(b) find $\mathrm{f}(x)$, writing your answer in simplest form.

(6)

(Total for Question 11 is 8 marks)

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Worked Solution - Question 11

Integration

Primary topic

1. Find point P

Since P has x-coordinate 4 and lies on the tangent $y = 3x + 4$, $P = (4, 16)$.

2. Find the normal

The tangent gradient is 3, so the normal gradient is $-\frac{1}{3}$. Hence the normal is $y - 16 = -\frac{1}{3}(x - 4)$.

3. Integrate f''(x)

$f''(x) = 4x + x^{-1/2}$, so $f'(x) = 2x^2 + 2\sqrt{x} + A$.

4. Use the tangent gradient

At $x = 4$, $f'(4) = 3$. Thus $2(16) + 2(2) + A = 3$, so $A = -33$.

5. Integrate again

$f(x) = \frac{2}{3}x^3 + \frac{4}{3}x^{3/2} - 33x + B$.

6. Use P(4, 16)

$16 = \frac{2}{3}(64) + \frac{4}{3}(8) - 33(4) + B = \frac{160}{3} - 132 + B$.

7. Find B

$B = \frac{284}{3}$, so $f(x) = \frac{2}{3}x^3 + \frac{4}{3}x^{3/2} - 33x + \frac{284}{3}$.

Final answer

(a) $y - 16 = -\frac{1}{3}(x - 4).$

(b) $f(x) = \frac{2}{3}x^3 + \frac{4}{3}x^{3/2} - 33x + \frac{284}{3}.$